

Rohit Singh

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Professional Summary

Software Engineer with 4+ years of experience designing and shipping backend systems, distributed data platforms, and serverless APIs. Strong expertise in **Python and Go** with proven experience building **high-throughput services** (sub-millisecond lookups across 2B+ records), leading platform modernization efforts, and delivering security-critical tooling under production incidents. Proven ownership from system design through production deployment.

Experience

Senior Associate Software Engineer, SonicWall – Remote Aug 2023 – Present

Telemetry Ingestion Service (Go · Microservices)

Built a **two-service streaming pipeline** (ingestor + store) in Go for collecting firewall telemetry from millions of devices. Implemented **zero-buffer multipart streaming**, spool-based async storage with drain workers, per-IP rate limiting, GeoIP resolution, and atomic file writes. Deployed as independently scalable containers with full CI/CD and load testing.

Credential Auditor API (Go · AWS · Bloom Filters)

Built a **serverless breach detection system** processing **2B+ credential hashes** using Bloom filters for **sub-millisecond lookups**. Implemented a dual-mode API (encrypted + pre-hashed workflows), bulk ingestion (16K-item batches), and secure **mTLS-based authentication**, on a containerized CI/CD pipeline with auto-scaling.

Security Incident Response Tool (Go · WASM)

Delivered a **zero-dependency browser-based** sanitization tool during a live security incident to remove sensitive data (VPN keys, credentials, RADIUS secrets) from firewall configurations. Built an **offline-capable WASM runtime** with a CLI remediation generator. **Recognized with a CEO spot award.**

Distributed Telemetry Processing Platform (Python · RabbitMQ)

Architected an **event-driven** telemetry system with multiprocessing workers, persistent **RabbitMQ queues**, retry mechanisms, and observability endpoints. Designed for **fault tolerance** and secure execution of telemetry queries via a **sandboxed runtime**.

Enterprise Migration Platform (FastAPI · React)

Rewrote a legacy Perl/CGI system into **FastAPI + React (TypeScript)** with **Redis-based session management**, multi-vendor configuration parsing, and end-to-end test coverage — improving modularity and deployment reliability.

TSR Analysis Platform (Python · Architecture)

Led decomposition of a monolithic tool into modular services. Extracted a reusable parser library with a **declarative model-driven architecture** (50+ typed models, 280+ auto-parsed fields), improving maintainability and enabling reuse across internal systems.

Associate Software Engineer, SonicWall – Bengaluru May 2022 – Aug 2023

- Maintained and extended enterprise migration tools for firmware compatibility across new device generations.
- Built automation utilities that reduced manual onboarding effort and improved system reliability.

Software Developer, Auton Systems Pvt. Ltd – Maharashtra Sep 2021 – May 2022

- Developed IoT and embedded systems using Raspberry Pi and ESP32.
- Worked on hardware–software integration for sensor-based systems and automation workflows.

Skills

Languages	Go, Python, JavaScript, TypeScript, C, Bash, Rust
Backend & API	Go (net/http), FastAPI, Django, Celery, REST APIs, WASM, Pydantic
Distributed Systems	Microservices, Event-Driven Architecture, RabbitMQ, Message Queues, Concurrency, Observability, System Design
Databases & DS	SQL (SQLite, PostgreSQL), NoSQL (MongoDB, Redis, Memcached), Bloom Filters
Cloud	AWS (ECS, Fargate, Lambda, KMS, SSM, ALB, ECR), Terraform
DevOps	Docker, Docker Compose, Nginx, GitLab CI/CD, Linux/Unix
Networking	TCP/IP, HTTP/HTTPS, DNS, IP Subnetting, NAT, Routing
Security	mTLS, Cryptography (AES/DES), OWASP, SAML, LDAP
Frontend	React, Vue.js, TypeScript, Tailwind CSS, Zustand, TanStack Query
AI Tooling	Claude Code, LLM-assisted debugging/development

Education

Mumbai University, B.Sc. in Computer Science — CGPA: 8.0/10.0 2018 – 2021

Certifications & Achievements

- **Spot Award by CEO** (Oct 2025) — Critical security tool delivery during company-wide incident
- **Certificate of Recognition** (Jan 2024) — Self-initiated platform architecture refactoring
- Developing Serverless Solutions on AWS | Building Data Lakes on AWS | AWS Well-Architected Best Practices